

Miles Hall

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EDUCATION

Loyola University Maryland - Baltimore, MD

Expected May 2027

Pursuing B.S. in Computer Science, minor in Finance and Data Science

Hyman Science Scholar | BGE Managed Scholar | GPA: 3.38

- **Honors:** Dean's List (Spring 2024, Fall 2025)

Relevant Coursework: Object-Oriented Data Structures, Object-Oriented Software Design, Computer Engineering

TECHNICAL SKILLS

Programming Languages: Java, JavaScript, TypeScript, Python, SQL, C, R, SAS, Bash/Zsh

Tools/Software: React, Next.js, Node.js, FastAPI, PostgreSQL, MongoDB, ChromaDB, Memgraph, Nomic, Docker, Git, AWS S3

WORK EXPERIENCE

Dell Technologies

Austin, TX

Software Engineering Intern

May 2026 – August 2026

- Built Oracle, a RAG platform indexing **300+** Dell CSG repositories across branches for unified natural-language code search.
- Engineered a hybrid retrieval pipeline combining **ChromaDB** vector search with **Memgraph** graph traversal to surface semantically relevant code, dependencies, and surrounding system context.
- Integrated **Amazon S3** and **MinIO** for multimodal ingestion and grounded answers traceable to source content.

Erickson Senior Living

Catonsville, MD

IT Training Team Intern

Jun 2025 – Present

- Manage employee access with MyEmma and resolve SharePoint, shared-drive, eLink, and MyEmma tickets through RemedyForce.
- Create onboarding documentation and support Microsoft 365 and Workday training to help employees adopt internal systems.

Headstarter

Virtual

Software Engineering Intern

Jun 2024 – Aug 2024

- Developed **5+** AI applications and APIs using Next.js, Pinecone, and Stripe API, reaching **500** users.
- Applied MVC architecture across five projects to separate interface, business, and data layers for maintainable iteration.

PROJECTS

CMDFlow (Python, FastAPI, React, MongoDB (Motor), Docker, Bash/Zsh)

Submission for HackPrinceton 2025

- Built a full-stack command-tracking platform using **FastAPI**, **MongoDB**, and a custom bash/zsh capture script to log, store, and organize real-time shell activity.
- Shipped JWT authentication, async ingestion, OpenAI-powered summaries, and instant search, transforming raw shell logs into a reusable development knowledge base.

Bucket (JavaScript, React, Firebase Firestore, Clerk, Gemini API, Next.js)

1st Place Overall & AI/ML Track Winner @ HackHounds 2025

- Engineered a Gemini-powered web application delivering personalized, location-aware recommendations with secure Clerk authentication and session management.
- Designed a timestamp-indexed Firestore schema for organized user-list queries and future usage analytics.

Studi (JavaScript, Next.js, React, Supabase, OpenAI/Gemini APIs)

- Built an AI study platform that transforms uploaded course materials into flashcards, practice tests, study guides, and grounded tutoring.
- Implemented spaced repetition with 1-60-day review intervals, mistake tracking, analytics, and authenticated Supabase sync to personalize study sessions.

LebronAI (JavaScript, Next.js, Groq, LLM Evaluation)

- Engineered a persona-driven chatbot with Next.js and Groq, adding rate-limit fallback routing and automated response revision for availability and character consistency.
- Developed an 18-case LLM evaluation suite and 19-example fine-tuning pipeline covering persona consistency, privacy, and prompt-injection resistance.

LEADERSHIP AND PROFESSIONAL DEVELOPMENT

Management Leadership for Tomorrow (Career Preparation Fellow)

Jan 2025 – Present

- Accepted into a selective 18-month professional development program for high-achieving diverse talent
- Complete business case studies and assignments to grow leadership and technical skills
- Attend conferences hosted by industry leaders, such as Deloitte, LinkedIn, and Target

ColorStack @ LUM (Vice President)

October 2024 - Present

- Led 7+ career-development and technical workshops for **30+** computing students, earning **95%** positive feedback and increasing participation and reported industry readiness by **50%**.